

A Power Model for D-MIMO Systems: Comparing the Energy Consumption of D-MIMO and Co-located MIMO

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Abstract—TODO
Index Terms—

I. INTRODUCTION

TODO small paragraph on energy consumption of 5G 6G networks

Recent works have focused on characterizing this energy consumption of wireless networks, and in particular of multiple input multiple output (MIMO) systems. In [1], the authors propose a detailed power consumption model for collocated MIMO systems, which is based on the power consumption of the different hardware components in the system for frequency range 3 (FR3) which is envisioned for 6G systems. Another prevelant enabler for 6G systems is distributed MIMO (D-MIMO)/cell-free MIMO. However, works that characterize the power consumption of D-MIMO systems is limited. TODO add som refs and work here Moreover, it is unclear whether the power consumption of D-MIMO systems is on par, lower or higher than the one of collocated MIMO systems. Especially, considering the additional network consumption that occurs from having multiple access point (AP), fronthaul links, synchronization, etc. In this work we introduce an extension of the power model introduced in [1] for the D-MIMO scenario. We incorporate TODO mention all extensions. TODO mention it is opensource an where it can be found.

II. D-MIMO SYSTEM MODEL

We consider a time division duplexing (TDD) D-MIMO network in which L APs, each with M antennas, jointly serve K single-antenna user equipments (UEs) over Q orthogonal frequency division multiplexing (OFDM) subcarriers, resulting in LM total antennas. All L APs are connected to a central processing unit (CPU) by fronthaul links. The indices $l = 1, \dots, L$, $m = 1, \dots, M$, $k = 1, \dots, K$ and $q = 1, \dots, Q$ denote the number of APs, antennas per AP, UEs and subcarriers respectively. Co-located massive MIMO (mMIMO) is the special case $L = 1$ with $M_{\text{ant}} = LM$ antennas on a single site at the centre of the coverage area. Co-located and distributed MIMO consider the same number of total antennas and radiate the same total power, they only differ in where the antennas sit.

We consider the same frame structure as in [1]. Namely, the total frame length is T_f which is split into a downlink (DL) and

uplink (UL) phase of length $\tau_{\text{DL}}T_f$ and $\tau_{\text{UL}}T_f$ respectively, with $\tau_{\text{DL}}, \tau_{\text{UL}} \in [0, 1]$. We consider N_{DL} available OFDM symbols in DL and N_{UL} in UL, of which $N_{\text{a,DL}}$ and $N_{\text{a,UL}}$ are activated and carrying data. The activated fraction is the average physical resource load

$$\bar{x}_i = \frac{N_{\text{a},i}}{N_i} \in [0, 1] \quad i \in \{\text{DL}, \text{UL}\}, \quad (1)$$

so $\bar{x}_i = 1$ is a fully loaded direction and $\bar{x}_i = 0$ one that carries reference signals only. In general, as defined in [1], the load also counts how many UEs are served and how many subcarriers each of them is allocated. We assume throughout that every activated symbol serves all K UEs on all Q subcarriers, so the load reduces to the ratio of activated to available symbols. The UL and DL frames are further subdivided into three parts as depicted in TODO add fig and ref it. The first part is for reference signaling and has length

$$\tau_i \tau_{\text{sig},i} \quad i \in \{\text{DL}, \text{UL}\}, \quad (2)$$

where $\tau_{\text{sig},i} \in [0, 1]$ is the fraction of the frame used for signaling. The second part is for data transmission and has length

$$\tau_i(1 - \tau_{\text{sig},i})\bar{x}_i \quad i \in \{\text{DL}, \text{UL}\}. \quad (3)$$

The third part represents a micro-sleep or discontinuous transmission (DTX) mode [2] and has length

$$\tau_i(1 - \tau_{\text{sig},i})(1 - \bar{x}_i) \quad i \in \{\text{DL}, \text{UL}\}. \quad (4)$$

A. Signal Model

Let $\mathbf{h}_{kl}[q] \in \mathbb{C}^M$ be the channel from AP l to UE k . Stacking the APs gives the collective channel $\mathbf{h}_k[q] = [\mathbf{h}_{k1}[q]^\top \dots \mathbf{h}_{kL}[q]^\top]^\top \in \mathbb{C}^{LM}$ and $\mathbf{H}[q] = [\mathbf{h}_1[q] \dots \mathbf{h}_K[q]] \in \mathbb{C}^{LM \times K}$.

1) *Downlink*: AP l transmits $\mathbf{x}_l[q] = \sum_k \mathbf{w}_{kl}[q]s_k[q]$, with $\mathbf{w}_{kl}[q] \in \mathbb{C}^M$ its precoding vector for UE k and $s_k[q]$ the unit-power data symbols, mutually independent across UEs. Let $\mathbf{w}_k[q] \in \mathbb{C}^{LM}$ be the collection of the L precoders of the APs $\mathbf{w}_{kl}[q]$, the signal received by UE k is

$$y_k[q] = \mathbf{h}_k[q]^H \mathbf{w}_k[q]s_k[q] + \sum_{i \neq k} \mathbf{h}_k[q]^H \mathbf{w}_i[q]s_i[q] + n_k[q] \quad (5)$$

thanks

with $n_k[q] \sim \mathcal{CN}(0, \sigma^2/Q)$. Treating the multi-user interference as noise,

$$\text{SINR}_k^{\text{DL}}[q] = \frac{|\mathbf{h}_k[q]^H \mathbf{w}_k[q]|^2}{\sum_{i \neq k} |\mathbf{h}_k[q]^H \mathbf{w}_i[q]|^2 + \sigma^2/Q}, \quad (6)$$

where σ^2 is the receiver noise power.

2) *Uplink*: UE k transmits with power $p_k[q] = p_k/Q$ subject to its own budget $p_k \leq p_{\max}$, and AP l observes

$$\mathbf{y}_l[q] = \sum_k \sqrt{p_k[q]} \mathbf{h}_{kl}[q] s_k[q] + \mathbf{n}_l[q] \quad (7)$$

with $\mathbf{n}_l[q] \sim \mathcal{CN}(\mathbf{0}, (\sigma^2/Q) \mathbf{I}_M)$. Applying a receive combiner $\mathbf{v}_k[q] \in \mathbb{C}^{LM}$ gives

$$\hat{s}_k[q] = \mathbf{v}_k[q]^H \mathbf{y}[q] = \sum_{l=1}^L \mathbf{v}_{kl}[q]^H \mathbf{y}_l[q]. \quad (8)$$

The resulting signal-to-interference-plus-noise ratio (SINR) is

$$\text{SINR}_k^{\text{UL}}[q] = \frac{p_k |\mathbf{v}_k[q]^H \mathbf{h}_k[q]|^2}{\sum_{i \neq k} p_i |\mathbf{v}_k[q]^H \mathbf{h}_i[q]|^2 + \sigma^2 \|\mathbf{v}_k[q]\|^2}. \quad (9)$$

B. Channel and Noise Model

Channels are drawn from the 3GPP TR 38.901 urban microcell model [3] through the Sionna implementation [4]. Each AP is a 38.901 base station carrying an M -element half-wavelength uniform linear array (ULA) of omnidirectional elements at height h_{AP} , and each UE is a single-antenna outdoor terminal at height h_{UE} . UEs are dropped uniformly over a square coverage area of side D , as are the APs. In the co-located case a single base station (BS) is placed at the centre. The receiver noise power follows from the thermal noise density, $\sigma^2 [\text{dBm}] = -174 + 10 \log_{10}(B/\text{Hz}) + F$, with B the bandwidth and F the noise figure in dB.

The model is specified through realizations rather than a closed-form correlation matrix, so the large-scale fading coefficient β_{kl} , used for power control, is estimated from the realization itself as

$$\beta_{kl} = \frac{1}{MQ} \sum_{q=1}^Q \|\mathbf{h}_{kl}[q]\|^2. \quad (10)$$

C. Centralized and Distributed Operation

Unlike the co-located case, which has a single sum-power budget, each AP has its own power constraint. Since the symbols are independent and unit-power, the power radiated by AP l over the OFDM block is

$$P_{T,l} = \sum_{q=1}^Q \sum_{k=1}^K \|\mathbf{w}_{kl}[q]\|^2 \leq \rho_{\max}. \quad (11)$$

How (11) is met is the difference between centralized and distributed operation, which is also related to the functional splits of Section III-A2.

1) *Centralized operation (splits S1, S2)*: The CPU holds $\mathbf{H}[q]$ and designs all precoders jointly. In the downlink it applies zero forcing (ZF),

$$\tilde{\mathbf{W}}[q] = \mathbf{H}[q] (\mathbf{H}[q]^H \mathbf{H}[q])^{-1} \in \mathbb{C}^{LM \times K}, \quad (12)$$

which nulls all inter-user interference and requires $K \leq LM$. A single power coefficient ρ_k is attached to each UE, since the precoding direction already fixes how that power spreads over the LM antennas. Fractional power control weights the UEs by their aggregate gain $\beta_k = \sum_l \beta_{kl}$,

$$\rho_k = P_{\text{tot}} \frac{\beta_k^v}{\sum_i \beta_i^v}, \quad P_{\text{tot}} = L \rho_{\max}, \quad (13)$$

with the exponent v trading throughput against fairness¹. This power normalization is applied as

$$\mathbf{w}_k[q] = \gamma \sqrt{\rho_k} \frac{\tilde{\mathbf{w}}_k[q]}{\sqrt{\sum_{q'} \|\tilde{\mathbf{w}}_k[q']\|^2}}, \quad \gamma = \sqrt{\frac{\rho_{\max}}{\max_l \tilde{p}_l}}, \quad (14)$$

where $\tilde{p}_l$ is the power AP l would radiate before that rescaling. Only the busiest AP then reaches its budget. Since $\sum_k \rho_k = L \rho_{\max}$ the nominal per-AP powers average to $\rho_{\max}$, so $\gamma \leq 1$, the constraint is always met. The uplink counterpart forwards the raw observations to the CPU, which combines over all LM antennas with $\mathbf{V}[q] = \mathbf{H}[q] (\mathbf{H}[q]^H \mathbf{H}[q])^{-1}$.

2) *Distributed operation (split S3)*: Each AP designs its own precoder/combiner from its local channel state information (CSI), so it suppresses only the interference it generates or observes itself and the coherent cross-AP cancellation is lost. Writing $\mathbf{E}_l[q] = [\mathbf{h}_{1l}[q] \cdots \mathbf{h}_{Kl}[q]] \in \mathbb{C}^{M \times K}$ for the local channel matrix and $\mathbf{P} = \text{diag}(p_1, \dots, p_K)$,

$$\mathbf{W}_l[q] = (\mathbf{E}_l[q] \mathbf{E}_l[q]^H + \lambda \mathbf{I}_M)^{-1} \mathbf{E}_l[q], \quad (15)$$

$$\mathbf{V}_l[q] = (\mathbf{E}_l[q] \mathbf{P} \mathbf{E}_l[q]^H + \lambda \mathbf{I}_M)^{-1} \mathbf{E}_l[q], \quad (16)$$

are the local regularized zero forcing (RZF) precoder/combiner which correspond to and the local minimum mean squared error (MMSE) precoder/combiner, with $\lambda = \sigma^2$. Each AP also allocates its own power budget among the UEs from local statistics,

$$\rho_{kl} = \rho_{\max} \frac{\beta_{kl}^v}{\sum_i \beta_{il}^v}, \quad (17)$$

so $\sum_k \rho_{kl} = \rho_{\max}$ by construction and every AP meets (11) with equality, $P_{T,l} = \rho_{\max}$, without any CPU-side rescaling. In the uplink the CPU fuses the L local estimates by summing them, $\hat{s}_k[q] = \sum_l \mathbf{v}_{kl}[q]^H \mathbf{y}_l[q]$.

In the uplink fractional power control is the same for both centralized and distributed operation at $p_k = \rho_{\max} \beta_k^{v_{\text{ul}}} / \max_i \beta_i^{v_{\text{ul}}}$. This is normalized by a maximum rather than a sum because each UE has its own independent power budget.

¹A positive v follows the channel and serves the UEs with the largest aggregate gain first, which maximizes the sum rate but starves those at the edge of the coverage area. A negative v compensates the weak UEs: at $v = -1$ the product $\rho_k \beta_k$ is the same for every UE, so all of them are received at approximately equal signal power, which is the fair extreme. Values in $[-1, 1]$ therefore interpolate between a throughput-optimal and a max-min-fair allocation.

D. Delivered Rates

The per-UE spectral efficiencies average over the subcarriers and over the Monte Carlo channel realizations,

$$SE_k^i = \tau_i (1 - \tau_{\text{sig},i}) \bar{x}_i \mathbb{E} \left(\frac{1}{Q} \sum_{q=1}^Q \log_2 (1 + \text{SINR}_k^i[q]) \right) \quad (18)$$

with $i \in \{\text{DL}, \text{UL}\}$, the prelog is the data fraction (3) of the frame, since reference signalling and the deactivated symbols carry no payload. The delivered sum rates can then be written as $R_i = \tilde{B} \sum_k SE_k^i$, with the effective bandwidth $\tilde{B} = 0.9 B$.

III. POWER MODEL DESCRIPTION

We first describe the consumption of a distributed network and outline the frame averaging, the functional split and the placement of the MIMO processing (Section III-A). Next, the two components that are specific to D-MIMO, i.e., the fronthaul and the synchronization of physically separate nodes (Section III-B) are described. Finally we describe the parametric model of the AP consumption, which is based on [1] (Section III-C).

TODO if no space can cut this cause a bit redundant. Also check if we keep load=1 or if we vary it in some sims The topology, the indices l, m, k, q , the per-AP budget $\rho_{\max}$ and radiated power $P_{T,l}$, the frame structure and the delivered rates $R_{\text{DL}}, R_{\text{UL}}$ are those of Section II, and $i \in \{\text{DL}, \text{UL}\}$ indexes denote the transmission directions. The APs are fully digital, so each carries M radio frequency (RF) chains. Every AP is assumed active and serves every user, and both directions are evaluated at full physical resource load, $\bar{x}_{\text{DL}} = \bar{x}_{\text{UL}} = 1$.

A. Network Power Consumption

The network consumption contains the L APs, the L fronthaul links that tie them to the CPU, and the CPU itself,

$$P_{\text{net}} = \sum_{l=1}^L \left(P_{\text{AP},l} + \frac{\overline{P_{\text{FH},l}}}{\eta_{\text{FH},\text{sc}}} \right) + \frac{\overline{P_{\text{CPU}}}}{\eta_{\text{CPU},\text{sc}}}, \quad (19)$$

where an overbar denotes a quantity averaged over one frame, as outlined in Section III-A1, and $\eta_{\text{FH},\text{sc}}, \eta_{\text{CPU},\text{sc}} \in (0, 1]$ are the supply-and-cooling efficiencies of the fronthaul transceivers and of the central unit. Each active AP retains the three-block structure of the co-located model from [1] and adds a constant synchronization consumption P_{sync} , detailed in Section III-B2,

$$P_{\text{AP},l} = \frac{\overline{P_{\text{dig},l}}}{\eta_{\text{dig},\text{sc}}} + \frac{\overline{P_{\text{ana},l}}}{\eta_{\text{ana},\text{sc}}} + \frac{\overline{P_{\text{PA},l}}}{\eta_{\text{PA},\text{sc}}} + \frac{P_{\text{sync}}}{\eta_{\text{sync},\text{sc}}}, \quad (20)$$

with the digital, analog and amplifier blocks detailed in Section III-C. The co-located baseline is recovered by setting $L = 1$, $M = M_{\text{ant}}$ and dropping the synchronization, fronthaul and CPU terms.

TABLE I: Node at which each digital operation runs. The uplink combiner follows the downlink precoder. Channel estimation is listed for completeness but is not priced, since perfect CSI is assumed. **TODO see what to do with channel estimation consumption**

Operation	S1	S2	S3
Encoding / decoding	CPU	CPU	CPU
Channel estimation	CPU	CPU	AP
Precoder computation	CPU	CPU	AP
Precoder application	CPU	AP	AP
OFDM, predistortion, BB filter	AP	AP	AP

1) *Frame structure and operating-mode averaging:* The overbars in (19) and (20) are averages over one frame, and every block of the power model is averaged over a frame. The frame structure is defined in Section II. The frame timing is shared by the whole network: under TDD the APs must switch direction together. In micro-sleep and idle a component drops to a fraction $\delta_c^\mu, \delta_c^0 \in [0, 1]$ of a base level P^0 . For a component c with working power P^a during data and P^s during signalling, the frame-averaged consumption of direction i is

$$\begin{aligned} \mathcal{A}_i^c(P^a, P^s, P^0) &= \bar{x}_i \tau_i (1 - \tau_{i,\text{sig}}) P^a + \tau_i \tau_{i,\text{sig}} P^s \\ &\quad + (1 - \bar{x}_i) \tau_i (1 - \tau_{i,\text{sig}}) \delta_c^\mu P^0 \\ &\quad + (1 - \tau_i) \delta_c^0 P^0. \end{aligned} \quad (21)$$

For the digital and analog blocks the three levels coincide, $P^a = P^s = P^0 = P$, and we write $\mathcal{A}_i^c(P) \triangleq \mathcal{A}_i^c(P, P, P)$; only the amplifier distinguishes them. Collecting the terms proportional to $\bar{x}_i$ separates each averaged consumption into a load-dependent and a load-independent part, which is what allows the consumption to be attributed to the delivered traffic.

2) *Functional split:* A single co-located site runs every digital operation in one place. A distributed network does not, and where each operation runs determines both the digital consumption and the fronthaul load. We consider the three splits of Table I. Split S1 forwards precoded samples: the CPU encodes, computes the centralized precoder and applies it, so AP l receives the M streams of frequency-domain samples. Split S2 forwards precoder coefficients and data: the CPU encodes and computes the precoder, but AP l applies it locally. Split S3 is fully local operation: the CPU only encodes, and each AP computes and applies its own precoder from local CSI. Splits S1 and S2 realize the same centralized precoder and therefore deliver the same rate; they differ only in where the work runs and in what crosses the fronthaul. S3 delivers a lower rate due to local processing.

3) *MIMO processing and its placement:* Following [1], a digital block consumes a static part plus its complex operations per sample Ξ evaluated at the constellation sampling rate $f_{s,I}$ and divided by the computational efficiency η of the implementation,

$$P_{\text{MIMO}} = \begin{cases} \bar{M} P_s + \frac{f_{s,I}}{\eta} \Xi, & \Xi > 0, \\ 0, & \Xi = 0, \end{cases} \quad (22)$$

TABLE II: Operations needed to *compute* the precoders and combiners of Section II-C, per subcarrier and per coherence block, derived in Appendix A. Dividing by v_{coh} gives the per-sample count Ξ_{comp} of (23). Applying the result costs a further $2K$ operations per sample and per processed antenna, for every scheme. **TODO maximum ratio (MR) is in the rate model but is not yet described in Section II-C; drop that row if it stays out of the paper.**

Scheme	Node	$v_{\text{coh}} \Xi_{\text{comp}}$
MR	any	0
ZF, RZF, MMSE	CPU	$\frac{K^3}{3} + 3K^2LM + KLM$
L-RZF, L-MMSE	AP l	$\frac{M^3}{3} + 3M^2K + MK$

with η the computational efficiency of the precoder in the downlink and of the combiner in the uplink, P_s the static power of one field-programmable gate array (FPGA) and $\bar{M}$ the number of FPGAs. Following [1], one FPGA per group of 32 RF chains: $\bar{M} = \lceil M/32 \rceil$ is used at an AP and $\bar{M} = \lceil LM/32 \rceil$ at the CPU, which processes the collective array. When considering S1, the AP has no precoder block at all, so $\Xi = 0$ and the AP has no idle FPGA.

Computing the precoder or combiner is considered separately from applying it to the signal, because the split moves these two operations between the AP and the CPU. Applying an already computed precoder or combiner to K UEs costs $2K$ operations per sample and per processed antenna whatever the scheme, hence $2KLM$ at the CPU and $2KM$ at one AP. Computing the precoder/combiner requires the scheme-dependent Ξ_{comp} of Table II, once per coherence block, and this cost is therefore spread over the v_{coh} samples of that block. We take the frame of Section II to span one coherence block, so the precoder is recomputed once per frame and the reference signalling fraction $\tau_{i,\text{sig}}$ is the share of that same block spent on pilots: one fading timescale governs both the precoder amortisation here and the signalling overhead of (18). The splits distribute the two counts over the two nodes as

$$\Xi_i^{\text{CPU}} = \iota_{S1} 2KLM + \iota_{S1,S2} \iota_{\text{DL}} \Xi_{\text{comp}}^{\text{cen}}, \quad (23)$$

$$\Xi_{i,l}^{\text{AP}} = \iota_{S2,S3} 2KM + \iota_{S3} \Xi_{\text{comp}}^{\text{loc}}, \quad (24)$$

where $\iota_{\bullet} \in \{0,1\}$ equals one for the split or direction named in its subscript. The centralized factorization is charged to the downlink only, since the ZF combiner of Section II-C is the same matrix as the ZF precoder and TDD reciprocity lets one factorization serve both directions. The local pair shares no factorization, because the uplink weights the local Gram matrix by the transmit powers $\mathbf{P}$ in (16) while the downlink does not, so under S3 the cost is charged in each direction.

4) *Central unit*: Channel encoding and decoding stay at the CPU under every split, since that is where the payload enters and leaves the network, and are driven by the delivered network sum rates R_{DL} and R_{UL} through the encoder and decoder models P_{enc} , P_{dec} of [1]. With the frame-averaging operator $\mathcal{A}_i$ of (21), the central unit consumes

$$\begin{aligned} \overline{P_{\text{CPU}}} &= P_{\text{CPU},0} + \mathcal{A}_{\text{DL}}(P_{\text{enc}} + P_{\text{MIMO,DL}}^{\text{CPU}}) \\ &\quad + \mathcal{A}_{\text{UL}}(P_{\text{dec}} + P_{\text{MIMO,UL}}^{\text{CPU}}), \end{aligned} \quad (25)$$

with $P_{\text{CPU},0}$ the always-on consumption of the central unit.

B. Fronthaul and Synchronization

1) *Fronthaul*: **TODO check the rate speed math** Following the cell-free power model of [5], the link serving AP l consumes a traffic-independent part covering the optical transceivers at both ends and a part proportional to the rate it carries,

$$P_{\text{FH},l} = P_{\text{FH},0} + \Pi_{\text{FH}} R_{\text{FH},l}, \quad (26)$$

with Π_{FH} in W per bit/s. What $R_{\text{FH},l}$ contains is set by the split of Section III-A2. Let b_{FH} be the number of bits per real sample, so a complex sample costs $2b_{\text{FH}}$ bits. Under S1 the link carries M streams of frequency-domain samples at the constellation rate in both directions and in both the data and the signalling mode,

$$R_{\text{FH},l}^{\text{S1}} = 2b_{\text{FH}} M f_{s,\text{I}}. \quad (27)$$

Under S2 the downlink carries the delivered payload plus the precoding coefficients, an $M \times K$ complex matrix per subcarrier refreshed once per coherence block,

$$R_{\text{FH},l}^{\text{S2,DL}} = R_{\text{DL}} + \frac{f_{s,\text{I}}}{v_{\text{coh}}} 2b_{\text{FH}} MK, \quad (28)$$

where R_{DL} appears in full because every AP serves every user. Under S3 only the payload is sent over the fronthaul links, $R_{\text{FH},l}^{\text{S3,DL}} = R_{\text{DL}}$. In the uplink, linear combining is distributive over the APs, $\hat{s}_k = \sum_l \mathbf{v}_{kl}^H \mathbf{y}_l$, so an AP holding its combining coefficients forwards K scalar partial sums instead of M sample streams under both S2 and S3,

$$R_{\text{FH},l}^{\text{S2,UL}} = R_{\text{FH},l}^{\text{S3,UL}} = 2b_{\text{FH}} K f_{s,\text{I}}. \quad (29)$$

The signalling mode separates the centralized splits from the local one. Centralized operation needs the CPU to see the raw received pilots in order to estimate the collective channel, so during the uplink signalling phase the link carries $2b_{\text{FH}} M f_{s,\text{I}}$ under S1 and S2 alike, whereas under S3 the pilots are consumed locally and the signalling load vanishes. Centralization therefore imposes a sample-level fronthaul during pilot transmission independent of the data-phase split.

The link is bidirectional and stays up across the whole frame, so its static term is charged once per link with a single duty cycle rather than once per direction. With

$$\theta_l = \sum_{i \in \{\text{DL,UL}\}} \tau_i [\tau_{i,\text{sig}} + (1 - \tau_{i,\text{sig}}) \bar{x}_i] \quad (30)$$

the fraction of the frame in which the link is in a working mode, and δ_{FH}^μ the reduction factor of an idle transceiver,

$$\overline{P_{\text{FH},l}} = [\theta_l + (1 - \theta_l) \delta_{\text{FH}}^\mu] P_{\text{FH},0} + \Pi_{\text{FH}} \overline{R_{\text{FH},l}}. \quad (31)$$

The averaged fronthaul rate reuses (21) with a zero base level, since an idle link carries no traffic,

$$\overline{R_{\text{FH},l}} = R_{\text{FH},l}^{\text{payload}} + \sum_{i \in \{\text{DL,UL}\}} \mathcal{A}_i(R_{\text{FH},l}^i, R_{\text{FH},l}^{i,\text{sig}}, 0), \quad (32)$$

where $R_{\text{FH},l}^{\text{payload}}$ is the delivered downlink payload of the data-sharing splits S2 and S3 and is excluded from the averaging, the remaining arguments then being peak rates. The reason is

that the map from rate to power in (26) is linear and $\mathcal{A}_i$ with a zero base is linear in its arguments, while the rate model already delivers R_{DL} with its prelog applied. Charging the delivered rate directly is thus equivalent to frame-averaging the corresponding peak rate, and doing both would count the frame structure twice. For the sample-forwarding split S1 the situation is the opposite: (27) is a hardware constant with no dependence on the delivered rate, and the frame structure is then the only mechanism that reduces the fronthaul load.

Remark 1 (Fronthaul ceiling on energy efficiency). *Under a data-sharing split the same payload is duplicated to every AP, so the network fronthaul load is $L R_{DL}$. Keeping only this term in (19) bounds the downlink energy efficiency by $1/(\Pi_{FH} L)$, independently of the bandwidth, the transmit power and the precoder. With $L = 16$ and $\Pi_{FH} = 0.25 \text{ W per Gbit/s}$ the network cannot exceed 250 Mbit J^{-1} however well everything else is designed, which is why the serving cluster size is the dominant lever on cell-free energy efficiency.*

2) *Synchronization*: Coherent joint transmission across physically separate nodes requires the APs to share a frequency and phase reference and requires the TDD chains to be reciprocity-calibrated. Whether the reference is distributed over the fronthaul, over a dedicated timing network, or over the air, the AP must hold it whether or not the frame is carrying data. It is therefore considered as the constant P_{sync} of (20), entirely load-independent and outside the frame averaging, with its own supply-and-cooling efficiency $\eta_{\text{sync,sc}}$.

What P_{sync} covers is the AP-side reference hardware only, that is the disciplined oscillator and the loop that slaves it to a reference. A shared master clock is not a per-AP cost and is counted in $P_{\text{CPU},0}$ instead. The over-the-air part of reciprocity calibration is frame time and CPU computation rather than a continuous per-AP power, and is not modelled. **TODO source a value for P_{sync} and $\eta_{\text{sync,sc}}$. The plausible range for the reference hardware spans roughly two orders of magnitude, from a temperature-controlled oscillator slaved over the fronthaul to an oven-controlled one with its own GNSS receiver, and at $L = 32$ that range moves P_{net} by about 18 %. Report it as a sensitivity band rather than a point value. Over-the-air calibration also consumes frame time, which is not yet charged to $\tau_{i,\text{sig}}$.**

C. Access Point and Co-located Power Consumption

The three frame-averaged blocks of (20) come from the FR3 BS model of [1], evaluated once per AP. A distributed AP carries few antennas, so hybrid beamforming has nothing to amortise and the array is fully digital, $M_{\text{RF}} = M_{\text{ant}} = M$ with no phase shifters. Substituting this into the analog and amplifier assemblies of [1] reduces them term for term to the per-AP expressions below. The digital block cannot be reused directly from [1] because the encoder and decoder move to the CPU and the MIMO processing terms depend on the functional split.

1) *Power amplifiers*: The amplifiers transmit only in the downlink. With output power p , the instantaneous consumption of one amplifier is

$$P_{\text{PA}}(p) = \frac{1}{\eta_{\text{PA,max}}} [\xi P_{\text{max}} + (1 - \xi) P_{\text{max}}^{1-\alpha} p^\alpha], \quad (33)$$

with P_{max} the maximum output power at a fixed back-off, $\eta_{\text{PA,max}}$ the maximum efficiency, $\xi \in [0, 1]$ the static-to-dynamic ratio and $\alpha \in [0, 1]$ the efficiency exponent. Each AP is an independent transmitter with its own budget ρ_{max} , so the single site power of [1] becomes the vector of radiated powers $P_{T,l}$ of (11), supplied by the rate model, which the two operation modes fill in differently: distributed operation meets the constraint with equality at every AP, $P_{T,l} = \rho_{\text{max}}$, whereas centralized operation rescales all APs by one common factor and brings only the busiest of them to its budget. Assuming the AP power is shared equally across its antennas, $P_{a,l} = P_{T,l}/M$, and

$$\overline{P_{\text{PA},l}} = M \mathcal{A}_{\text{DL}}^{\text{PA}}(P_{\text{PA}}(P_{a,l}), P_{\text{PA}}(\zeta_{\text{sig}} P_{\text{max}}), P_{\text{PA}}(0)), \quad (34)$$

with ζ_{sig} the fraction of P_{max} used during signalling. The equal split is not exact, since the per-antenna powers differ for any non-isotropic precoder, but because $\alpha \in [0, 1]$ makes $P_{\text{PA}}(\cdot)$ concave, Jensen's inequality gives $\frac{1}{M} \sum_m P_{\text{PA}}(P_{a,lm}) \leq P_{\text{PA}}(P_{a,l})$: it is a conservative upper bound, tight when the precoder loads an AP uniformly.

Remark 2 (Amplifier sizing). *The relation between P_{max} and the budget decides how the static term scales with the deployment and must be stated. We size each amplifier for its share of the budget it is actually given, $P_{\text{max}} = \rho_{\text{max}}/M$ at an AP and $P_{\text{max}} = P_{\text{TX}}/(LM)$ at the co-located site. Since the comparison holds the total transmit power fixed, $\rho_{\text{max}} = P_{\text{TX}}/L$, the two deployments then carry amplifiers of identical rating and differ only in how many sites they sit on. Substituting (33) into (34) makes the data-mode consumption of an AP independent of M ,*

$$M P_{\text{PA}}(P_{a,l}) = \frac{\rho_{\text{max}}}{\eta_{\text{PA,max}}} \left[\xi + (1 - \xi) \left(\frac{P_{T,l}}{\rho_{\text{max}}} \right)^\alpha \right], \quad (35)$$

*so the amplifiers see the precoders of Section II-C only through the fraction $P_{T,l}/\rho_{\text{max}} \in [0, 1]$ of the budget that AP l radiates, and the network static floor is $L\xi\rho_{\text{max}}/\eta_{\text{PA,max}} = \xi P_{\text{TX}}/\eta_{\text{PA,max}}$, the same for both deployments at equal total transmit power. The per-AP constraint then costs the network twice under centralized operation: only part of the budget is radiated, while the static floor is paid in full. **TODO source $\xi, \alpha, \eta_{\text{PA,max}}$ for AP-class amplifiers; the values used are fitted for macro-cell hardware.***

2) *Analog front end*: The analog block comprises, per RF chain, the digital-to-analog converters (DACs) in the downlink or analog-to-digital converters (ADCs) in the uplink, RF filters, mixers and the low-noise amplifiers, with the local oscillator

shared across the chains of one node. Specialising [1] to a fully digital array without phase shifters,

$$P_{\text{ana,DL},l} = P_{\text{LO}} + M(2P_{\text{DAC}} + 2P_{\text{filt,RF}} + 2P_{\text{mix}}), \quad (36)$$

$$P_{\text{ana,UL},l} = P_{\text{LO}} + M(2P_{\text{ADC}} + 2P_{\text{filt,RF}} + 2P_{\text{mix}} + P_{\text{LNA}}), \quad (37)$$

the factor two accounting for the in-phase and quadrature branches, and

$$\overline{P_{\text{ana},l}} = \mathcal{A}_{\text{DL}}^{\text{ana}}(P_{\text{ana,DL},l}) + \mathcal{A}_{\text{UL}}^{\text{ana}}(P_{\text{ana,UL},l}) + P_{\text{ana,misc}}. \quad (38)$$

This is the co-located analog block term for term, with no distributed addition: the synchronization of Section III-B2 sits outside it, in (20). The local oscillator is the first term to change character: a co-located array shares one across all M_{ant} chains, whereas the distributed network pays $L P_{\text{LO}}$.

3) *Digital signal processing*: Every AP performs the OFDM modulation and demodulation, the predistortion of its own amplifiers and the baseband filtering whatever the split, so with the component models P_{IFFT} , P_{DPD} and $P_{\text{filt,BB}}$ of [1] evaluated at M chains,

$$P_{\text{dig,DL},l} = P_{\text{IFFT}} + P_{\text{DPD}} + P_{\text{filt,BB}} + P_{\text{MIMO,DL},l}, \quad (39)$$

$$P_{\text{dig,UL},l} = P_{\text{IFFT}} + P_{\text{filt,BB}} + P_{\text{MIMO,UL},l}, \quad (40)$$

with $\overline{P_{\text{dig},l}} = \mathcal{A}_{\text{DL}}^{\text{dig}}(P_{\text{dig,DL},l}) + \mathcal{A}_{\text{UL}}^{\text{dig}}(P_{\text{dig,UL},l})$. The encoder and decoder are absent by construction, which is the structural difference from the single-site model, and only the MIMO term of (22) depends on the split.

Remark 3 (Loss of static-power amortisation). *The precoder, combiner and baseband filter share a static power P_s per FPGA, one per group of 32 chains. A co-located array of M_{ant} antennas amortises that power over 32 chains at a time, whereas a network of L APs with $M < 32$ antennas each pays it L times. At $L = 16$ and $M = 4$ the network runs 16 FPGAs where a co-located array of the same 64 antennas runs 2. Chopping a large array into many small ones leaves the per-chain digital costs untouched but destroys the sharing of the fixed ones, and this is a general feature rather than an artefact of the value 32.*

4) *Co-located baseline*: The reference deployment is the unmodified model of [1] evaluated at a single site: one fully digital array of $M_{\text{ant}} = LM$ antennas radiating the full budget P_{TX} , one shared local oscillator, no fronthaul, no central unit and no synchronization power, with the complete digital chain (encoder, precoder, OFDM, predistortion and baseband filter in the downlink; decoder, combiner, OFDM and baseband filter in the uplink) at that one site.

D. Delivered Rate and Energy Efficiency

The delivered sum rates R_{DL} and R_{UL} come from the rate model of Section II with the data-fraction prelogs of (18) already applied, and the network energy efficiency is the delivered rate per consumed watt,

$$\text{EE}_{\text{net}} = \frac{R_{\text{DL}} + R_{\text{UL}}}{P_{\text{net}}} \quad [\text{bit/J}], \quad (41)$$

which reduces to the single-site energy efficiency of [1] for the co-located baseline. Because a distributed and a co-located deployment cover the same area with the same hardware count, the comparison is made at equal LM and equal total transmit power.

IV. NUMERICAL EVALUATION

A. Parameters

The evaluation needs three groups of parameters. The deployment, channel and frame of Section II are collected in Table III. The hardware constants of the AP and of the co-located baseline are inherited unchanged from the FR3 BS model of [1] and are listed in Table IV, so that the two deployments cannot differ in a hardware constant. The quantities that exist only in a distributed network, that is the fronthaul, the central unit and the per-AP synchronization, are collected in Table V.

TODO Parameters that still need a decision rather than a source. (i) The headline topology: Remark 1 and Remark 3 quote $L = 16$, $M = 4$, whereas the rate model is configured with $L = 40$, $M = 4$, $K = 20$; one of the two has to move. (ii) The frame in Table III gives $\tau_{\text{UL,sig}} = 1$, so the uplink phase carries nothing but pilots and $R_{\text{UL}} = 0$. The evaluation is then downlink-only and (41) reduces to $R_{\text{DL}}/P_{\text{net}}$; either say so explicitly or give the uplink a data phase. (iii) With $\tau_{\text{DL,sig}} = 0$ the amplifier signalling level of (34) is never reached, so ζ_{sig} has no effect on the reported numbers. (iv) v_{coh} is stated outright as 200 samples; the Doppler-derived value of [1], at a maximum path-length difference of 300 m and a receiver velocity of 30 m s^{-1} , would be about 714 samples at $f_c = 7 \text{ GHz}$. Pick one and state which. (v) $Q = 64$ subcarriers are evaluated while the OFDM system carries $Q_{\text{IFFT}} = 4096$; say that the former is a Monte Carlo subset so the two are not read as the same quantity. (vi) (22) charges $P_s = 1 \text{ W}$ per FPGA to the precoder, the combiner and the baseband filter, while the IFFT and predistortion blocks each carry a flat 0.1 W that does not depend on the antenna count. Remark 3 discusses only the first, but both are paid once per node: the network pays $L \times 0.2 \text{ W}$ of IFFT and predistortion static where the co-located site pays 0.2 W, an exactly L -fold penalty, against the $L[M/32]$ versus $[LM/32]$ ratio of the FPGA term. The remark therefore understates the loss of amortisation; either extend it or say that it treats the dominant term only.

V. CONCLUSION

TODO future work:

- include channel estimation error in rate calculation
- include channel estimation power consumption in power model
- include pilot contamination in rate calculation
- no AP selection, all APs are active, investigate the impact on energy efficiency
- make some parameters more realistic e.g. fronthaul power consumption, synchronization power consumption, etc.
- investigate the impact of hardware impairments on system performance

TABLE III: Deployment, channel and frame parameters of Section II. Entries marked ‡ are design choices that still have to be fixed; the remaining rows are either cited or derived from a cited quantity. The noise figure is an assumption rather than a measured value, and σ^2 inherits that status through Section II-B. **TODO settle every ‡ row before the figures are final; the topology row in particular, since Remark 1 and Remark 3 quote $L = 16$, $M = 4$ while the rate model is configured with $L = 40$, $M = 4$, $K = 20$.**

Symbol	Description	Value
<i>Topology</i>		
L	Number of APs	16 ‡
M	Antennas per AP	4 ‡
LM	Total antennas, both deployments	64
K	Single-antenna UEs	20 ‡
Q	Subcarriers evaluated	64 ‡
D	Coverage-area side	200 m ‡
h_{AP}	AP height [3]	10 m
h_{UE}	UE height [3]	1.5 m
<i>Radio</i>		
f_c	Carrier frequency	7 GHz ‡
B	Bandwidth [1]	400 MHz
Δ_f	Subcarrier spacing [1]	120 kHz
$\tilde{B}$	Effective bandwidth, $0.9B$	360 MHz
F	Receiver noise figure	8 dB ‡
σ^2	Noise power, from Section II-B	-80.0 dBm
<i>Transmit power and power control</i>		
ρ_{max}	Per-AP budget	1 W ‡
P_{TX}	Total budget, $L\rho_{\text{max}}$	16 W
p_{max}	Per-UE uplink budget	0.1 W ‡
v	Downlink fractional exponent	0.5 ‡
v_{ul}	Uplink fractional exponent	-0.5 ‡
λ	Local RZF loading	σ^2 ‡
<i>Frame</i>		
τ_{DL}	Downlink share of the frame	0.9 ‡
τ_{UL}	Uplink share, $1 - \tau_{\text{DL}}$	0.1
$\tau_{\text{DL},\text{sig}}$	Downlink signalling fraction	0 ‡
$\tau_{\text{UL},\text{sig}}$	Uplink signalling fraction	1 ‡
$\bar{x}_{\text{DL}}, \bar{x}_{\text{UL}}$	Resource load	1 ‡
v_{coh}	Samples per coherence block	200 ‡
<i>Monte Carlo</i>		
	Channel realizations per point	100 ‡
	Channel model [3], [4]	38.901 UMi

- accurately characterize all parameters based on the size of the BS/APs.

APPENDIX A

OPERATION COUNTS OF THE PRECODERS AND COMBINERS

This appendix derives the entries of Table II. Operations are counted as in [1]: a complex multiply-accumulate is two operations, so an inner product of two length- r complex vectors costs $2r$ operations and the efficiencies η of (22) are expressed per operation. All counts below are per subcarrier and per coherence block; dividing by v_{coh} turns them into the per-sample figures used in (23). The subcarrier index is dropped throughout.

A. Computing the precoder or combiner

Every scheme of Section II-C other than MR has the same three stages, and the two families differ only in which

dimension is inverted. Let $\mathbf{A} \in \mathbb{C}^{n \times n}$ be the Hermitian positive definite matrix that is factorized and let r be the length of the vectors from which its entries are built, which is also the number of right-hand sides that are solved against it. The centralized precoder (12) and its uplink counterpart form $\mathbf{A} = \mathbf{H}^H \mathbf{H}$ and solve $\tilde{\mathbf{W}} \mathbf{A} = \mathbf{H}$ for the LM rows of $\mathbf{H} \in \mathbb{C}^{LM \times K}$, so $n = K$ and $r = LM$. The local schemes (15) and (16) form $\mathbf{A} = \mathbf{E}_l \mathbf{E}_l^H + \lambda \mathbf{I}_M$ and solve $\mathbf{A} \mathbf{W}_l = \mathbf{E}_l$ for the K columns of $\mathbf{E}_l \in \mathbb{C}^{M \times K}$, so $n = M$ and $r = K$. The three stages cost

- 1) *Gram matrix*. $\mathbf{A}$ is Hermitian, so only its $n(n+1)/2$ upper-triangular entries are formed, each an inner product of length r at $2r$ operations, giving $(n^2 + n)r$. The diagonal loading $\lambda \mathbf{I}_n$ adds n real additions, and the power weighting $\mathbf{P}$ of (16) rescales the K columns of $\mathbf{E}_l$ beforehand at $O(MK)$; both are dropped as lower order.
- 2) *Cholesky factorization*. $\mathbf{A} = \mathbf{L} \mathbf{L}^H$ costs $n^3/3$ operations, the count used in [1].
- 3) *Triangular solves*. Forward and backward substitution against $\mathbf{L}$ costs n^2 complex multiply-accumulates per right-hand side [6, App. B.1.1], hence $2n^2$ operations, and there are r of them, giving $2n^2 r$.

Summing the three stages,

$$v_{\text{coh}} \Xi_{\text{comp}}(n, r) = \frac{n^3}{3} + 3n^2 r + n r, \quad (42)$$

which is the second row of Table II at $(n, r) = (K, LM)$ and the third at $(n, r) = (M, K)$. Setting r to the number of RF chains of a single site recovers the co-located count of [1] exactly, so the distributed and co-located deployments are priced by the same expression and can be compared. The MR row costs nothing to compute: under the perfect CSI assumed here the direction is the channel itself, $\tilde{\mathbf{w}}_k = \mathbf{h}_k$ in the downlink and $\mathbf{v}_{kl} = \mathbf{h}_{kl}$ in the uplink, so no factorization is performed.

Only the middle term of (42) is a genuine cubic cost in the antenna count, and it is the one that separates the two families. At $(n, r) = (K, LM)$ the factorization grows as $K^2 LM$ and is carried by a single node, whereas at $(n, r) = (M, K)$ each of the L APs pays $M^2 K$, for a network total of $LM^2 K$. Local processing is therefore cheaper than centralized processing by a factor K/M in this term, which is the computational counterpart of the rate loss it accepts.

Remark 4 (Convention on the cubic term). *Counting a complex multiply-accumulate as two operations, as stages 1 and 3 do, would give $2(n^3 - n)/3$ rather than $n^3/3$ for the Cholesky factorization [6, App. B.1.1]. We keep $n^3/3$ so that (42) reproduces [1] term for term and the co-located baseline is not priced by a different rule than the distributed network. The residual ambiguity is the ratio $n/(9r)$ of the factorization to the dominant $3n^2 r$ term, which is small wherever the matrix is small relative to the number of right-hand sides. Both families satisfy this at the operating point used here, $K = 20$ UEs against $LM = 64$ antennas centrally and $M = 4$ antennas*

TABLE IV: Hardware constants of the AP and of the co-located baseline, inherited unchanged from the FR3 BS model of [1] and applied to both deployments. P_s is the static power of one FPGA and is charged to the precoder, the combiner and the baseband filter; the encoder, decoder, inverse fast Fourier transform (IFFT) and predistortion blocks carry a further 0.1 W each inside the component models of that reference. $P_{\max}$ follows the sizing convention of Remark 2 rather than being a component rating. The phase-shifter figure of merit is not listed because the array is fully digital. **TODO units are missing for $\Xi_{\text{DAC},1}$, $\Xi_{\text{DAC},2}$, Ξ_{LNA} and Ξ_{mix} ; take them from [1] or drop the rows, since a bare number is not reproducible. ξ , α and $\eta_{\text{PA},\max}$ were fitted for macro-cell amplifiers and should be refitted for AP-class hardware (Remark 2). Every value here is traceable to the authors' own evaluation script, but the equation numbering that script is documented against does not match [1], so confirm each value against the published version before quoting it.**

Symbol	Description	Value	Symbol	Description	Value
<i>Supply and cooling</i>			<i>Analog front end</i>		
$\eta_{\text{dig},\text{sc}}$	Digital block	0.81	b_{DAC}	DAC effective bits	8
$\eta_{\text{ana},\text{sc}}$	Analog block	0.81	f_{DAC}	DAC sampling frequency	5 GHz
$\eta_{\text{PA},\text{sc}}$	Amplifier block	0.81	$\Xi_{\text{DAC},1}$	DAC figure of merit 1	1.5×10^{-5}
<i>Power amplifier</i>			$\Xi_{\text{DAC},2}$	DAC figure of merit 2	1.5×10^{-12}
ξ	Static-to-dynamic ratio	0.1	b_{ADC}	ADC effective bits	8
α	Efficiency exponent	0.75	f_{ADC}	ADC sampling frequency	5 GHz
$\eta_{\text{PA},\max}$	Maximum efficiency	0.15	W_{ADC}	ADC Walden figure of merit	70 fJ
$P_{\max}$	Rating, $\rho_{\max}/M$	0.25 W	Ξ_{LNA}	low-noise amplifier (LNA) figure of merit	2.7×10^{-11}
ζ_{sig}	Signalling power fraction	1/12	Ξ_{mix}	Mixer figure of merit	2.5×10^{-13}
<i>Digital processing</i>			P_{LO}	Local oscillator	40 mW
Q_{IFFT}	IFFT size	4096	$P_{\text{filt},\text{RF}}$	RF filter	5 mW
μ	Sampling-frequency factor	0.9	$P_{\text{ana},\text{misc}}$	Miscellaneous analog	0 W
$f_{s,\text{I}}$	Constellation rate, μB	360 MHz	<i>Computational efficiencies [OPS/W]</i>		
$f_{s,\text{II}}$	Baseband rate, $Q_{\text{IFFT}}\Delta_f$	491.52 MHz	$\eta_{\text{enc}}, \eta_{\text{dec}}$	Encoder, decoder (ASIC)	2×10^{12}
Ξ_{DPD}	Predistortion, ops per sample	50	$\eta_{\text{IFFT}}, \eta_{\text{DPD}}$	IFFT, predistortion (ASIC)	2×10^{12}
$n_{\text{filt},\text{BB}}$	Baseband-filter taps	20	η	Precoder, combiner (FPGA)	0.2×10^{12}
$o_{\text{filt},\text{BB}}$	Baseband oversampling	4	$\eta_{\text{filt},\text{BB}}$	Baseband filter (FPGA)	0.2×10^{12}
P_s	Static power per FPGA	1 W			
	RF chains per FPGA	32			
<i>Reduction factors (micro-sleep / idle)</i>					
$\delta_{\text{dig}}^\mu, \delta_{\text{dig}}^0$	Digital	0.5, 0.25			
$\delta_{\text{ana}}^\mu, \delta_{\text{ana}}^0$	Analog	0.75, 0.5			
$\delta_{\text{PA}}^\mu, \delta_{\text{PA}}^0$	Amplifier	0.5, 0.25			

against $K = 20$ UEs locally, giving 3.5 % and 2.2 %. It ceases to hold for the local schemes once an AP carries as many antennas as the network serves UEs, so a deployment with few large APs should be checked against the doubled term. **TODO fix these numbers once K and the deployment sweep are final.**

B. Applying the precoder or combiner

Applying is scheme-independent. AP l forms $\mathbf{x}_l = \mathbf{W}_l \mathbf{s}$ with $\mathbf{W}_l \in \mathbb{C}^{M \times K}$, which is MK complex multiplications and $M(K-1)$ complex additions, hence $2KM$ operations per sample to leading order. In the uplink the partial sums $\mathbf{v}_{kl}^H \mathbf{y}_l$ of (8) cost the same, K inner products of length M . A node that processes all LM antennas, which is the CPU under S1, pays $2KLM$ instead. This is the $2K$ operations per sample and per processed antenna quoted in Section III-A3 and it appears once per direction, since the precoded samples and the combined samples are distinct data.

C. What is not counted

The power control of (13) and (17) needs the precoder norms $\sum_q \|\tilde{\mathbf{w}}_k[q]\|^2$ and the large-scale coefficients (10), which is $O(KLM)$ per coherence block across the network and therefore below the $O(K^2LM)$ factorization it is attached to. Channel estimation is not priced either, since perfect CSI

is assumed, and Table I lists it for completeness only. Neither omission is specific to the distributed deployment, so both cancel in the comparison of Section III-D.

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TABLE V: Parameters specific to the distributed network. A value is given only where it comes from a reference; TBD marks a quantity that the co-located model of [1] does not contain and for which no source has been established, so no number is quoted here. **TODO source every TBD row. The implementation currently runs placeholders for them, chosen only to be of the right order of magnitude; they are deliberately not printed here, and any conclusion that turns on one of them has to be reported as a sensitivity range rather than a point value. Check $P_{\text{FH},0}$ and Π_{FH} against [5] before quoting them.**

Symbol	Description	Value	Source
<i>Fronthaul</i>			
$P_{\text{FH},0}$	Traffic-independent power per link	0.825 W	[5]
Π_{FH}	Traffic-dependent coefficient	0.25 W/Gbit/s	[5]
b_{FH}	Bits per real sample	TBD	
δ_{FH}^{μ}	Idle transceiver reduction	TBD	
$\eta_{\text{FH},\text{sc}}$	Supply and cooling	TBD	
<i>Central unit</i>			
$P_{\text{CPU},0}$	Always-on consumption	TBD	
$\eta_{\text{CPU},\text{sc}}$	Supply and cooling, $\approx 1/\text{PUE}$	TBD	
	RF chains per CPU FPGA	32	[1]
<i>Synchronization</i>			
P_{sync}	Per-AP reference hardware	TBD	
$\eta_{\text{sync},\text{sc}}$	Supply and cooling	TBD	